The Octonionic Round Trip and Its Genuine Total

Insertion outline for the Concentricity proof

Jesse Michael Paul

24 July 2026

Purpose. This is a structural insertion outline, not final exposition. It belongs after the slice-wise construction of the distinguished Euler–Weierstrass–GPV action and before the application of Riehl 8.3.4, π_0, and val. Its purpose is to make explicit the arrow-level passage from the slice geometry to the genuine total of the octonionic round trip.

1 The three categorical levels

1.1 The slice projection

The slice-wise geometric action is the functor

$$A_{\text{slice}}\colon \text{GreatCircle.Base} \longrightarrow \text{SphereWorld}.$$

It contains the two-point stabilizer, the unique orbit–stabilizer factorization, the full prime-indexed Euler lift, its GPV winding data, and the pointwise tame continuous triangles to N. It is essential, but it is the sphere/Moebius projection of the action rather than the octonionic round trip itself.

1.2 The octonionic round-trip functor

The native home of the A-section is $\mathbb{O}^*$. Its actual round trip is the G_2-equivariant endofunctor

$$A_{\mathbb{O}}\colon H_1 \longrightarrow H_1, \qquad s \longmapsto A.\operatorname{realize}(s).$$

The normalized state world has natural input and output functors, and the round trip is their factorization:

$$\begin{tikzcd} \text{StateWorld}_A \arrow[r, "\text{In}"] \arrow[dr, "\text{Out}"'] & H_1 \arrow[d, "A_{\mathbb{O}}"] \\ & H_1 \end{tikzcd} \qquad \text{Out} = A_{\mathbb{O}} \circ \text{In}\,.$$

This triangle is arrow-level data. It says that the input and output move naturally under the same G_2 direction transport; it is stronger than the object assignment $s \mapsto A.\operatorname{realize}(s)$.

1.3 The projective indexing diagram

The complete normalized state and its analytic presentation are then curried over the projective base:

$$\mathscr{A}_A\colon \text{GreatCircle.Base} \longrightarrow \mathbf{Grpd}.$$

For a projective object X,

$$\mathscr{A}_A(X) = \text{StateWorld}_A \times \text{PresentationWorld}_A(X).$$

The first factor contains the physical input/output round trip. The second contains the full Euler–Weierstrass–GPV tape and all of its pointwise triangles to N. Thus $\mathscr{A}_A$ is the Grothendieck-indexing diagram of the round-trip geometry, not a second octonionic endofunctor.

2 The distinguished element and the two sweeps

2.1 Vertical: the entire prime stack through N

For every Euler path δ the lift is literally

$$L_\delta(t) = \sum\nolimits'_{p:A.\iota} A.\ell_p(\delta(t)).$$

The prime index remains internal to the infinite sum. At every instant,

$$\begin{tikzcd}
L_\delta(t) \arrow[r, "\exp"] \arrow[d, rightsquigarrow, "\text{unique tame lift}"'] & A.F(\delta(t)) \arrow[d, "\text{diagonal Moebius action}"] \\
N\text{-positioned lift} \arrow[r, "\text{same action at } N"'] & N\text{-positioned value}
\end{tikzcd}$$

commutes. The real part of the lift is the level; its imaginary part is the winding. Euler and Weierstrass are two presentations of this one element.

2.2 Horizontal: unique orbit–stabilizer factorization

Every projective arrow has the unique factorization

$$f = \mathrm{orbitRep}(Y)\ \mathrm{stab}(f)\ \mathrm{orbitRep}(X)^{-1}.$$

At every instant of every GPV tape it supplies a two-legged commuting square:

$$\begin{tikzcd}
\text{input action at } X \arrow[r, "D_X(t)"] \arrow[d, "\mathrm{right}(f{,}t)"'] & \text{A-output at } X \arrow[d, "\mathrm{left}(f{,}t)"] \\
\text{input action at } Y \arrow[r, "D_Y(t)"'] & \text{A-output at } Y.
\end{tikzcd}$$

In multiplicative form,

$$\mathrm{left}(f,t)\, D_X(t) = D_Y(t)\ \mathrm{right}(f,t).$$

The winding lies in the stabilizer factor. The orbit factor performs the horizontal sweep. Identity and composition of these squares are inherited from uniqueness of orbit–stabilizer factorization.

3 The genuine total is a total of the round trip

The underlying Grothendieck carrier is

$$\mathcal{T}_A = \int_{\mathrm{GreatCircle.Base}} \mathscr{A}_A.$$

An object is a projective base point together with the complete normalized physical/presentation state. A morphism consists of a base arrow and a fibre arrow after the corresponding projective reindexing.

This carrier becomes the genuine total of the A-section only when the input and output are functors on *total arrows*:

$$\mathrm{In}_{\mathcal{T}_A}, \mathrm{Out}_{\mathcal{T}_A} : \mathcal{T}_A \longrightarrow H_1,$$

and the octonionic round trip commutes on the total:

$$\begin{tikzcd} \mathcal{T}_A \arrow[r, "\mathrm{In}_{\mathcal{T}_A}"] \arrow[dr, "\mathrm{Out}_{\mathcal{T}_A}"'] & H_1 \arrow[d, "A_{\mathbb{O}}"] \\ & H_1. \end{tikzcd} \qquad \mathrm{Out}_{\mathcal{T}_A} = A_{\mathbb{O}} \circ \mathrm{In}_{\mathcal{T}_A}.$$

For a total morphism $\langle f, \varphi \rangle$, the fibre morphism φ occurs after $\mathscr{A}_A(f)$. The projective/GPV action square is precisely the compatibility that makes $\mathrm{In}_{\mathcal{T}_A}$ and $\mathrm{Out}_{\mathcal{T}_A}$ functorial on this composite arrow.

Why this is the missing gate. An object function $\mathrm{Ob}(\mathcal{T}_A) \to \mathrm{Ob}(H_1)$ can store output values, but it cannot express naturality. A cocone can be natural only on a functor. Therefore the output cocone is native to the construction only after $\mathrm{Out}_{\mathcal{T}_A}$ acts on arrows and the triangle above is proved.

4 The natural $9 + 3$ partition

The twelve readings are not installed as twelve fields and are not reintroduced after totalization.

Nine action readings	**Three codomain/output readings**
Euler level, the complete GPV lift, endpoint real-level conservation, winding, complex exponential fibre level and height, the octonionic chart, and orbit–stabilizer transport.	The residue-$\mathbb{C}$ sphere population, its infinitude, and the intrinsic real coordinate of each realized sphere.

The nine readings spread through $\mathscr{A}_A$ by the certified action squares. The three output readings occur in the functorial output $\mathrm{Out}_{\mathcal{T}_A}$. Their partition is therefore a consequence of one equivariant total, not a later conjunction of hypotheses.

5 The output population and the boundary before 8.3.4

For every index n and sphere world I, the A-action produces a total object carrying together:

- the normalized octonionic point on the residue-$\mathbb{C}$ sphere;
- its value under $A_{\mathbb{O}}$;
- its intrinsic real register;
- the complete Euler–Weierstrass–GPV presentation;
- its native orbit–stabilizer branch to the N-positioned state.

No common value at N is included in the carrier. Before the next theorem is applied, the proof has established only the following:

$$\{\text{A-generated residue spheres}\} \hookrightarrow \mathcal{T}_A \underset{\mathrm{Out}_{\mathcal{T}_A}}{\overset{\mathrm{In}_{\mathcal{T}_A}}{\rightrightarrows}} H_1.$$

The family is now arrow-natural and output-bearing. Its common component and common real centre remain conclusions.

6 Only now: the categorical readout

After the total-level triangle is kernel-green, the existing machinery is applied in this order:

$$\mathcal{T}_A \xrightarrow{\pi_0} \pi_0(\mathcal{T}_A) \xrightarrow[\sim]{\text{Riehl 8.3.4}} \operatorname*{colim}_X \pi_0\big(\mathscr{A}_A(X)\big) \longmapsto \kappa \xrightarrow{\text{val}} c \in \mathbb{R}.$$

Here the component diagram, its cocone, quotient representatives, and descent are the internal categorical machinery specialized to the exact total. They are not predeclared replacements. The atomic conclusion is

$$\exists c \in \mathbb{R}, \quad \forall n \in \mathbb{N}, \quad \mathrm{Re}(A.\,\mathrm{sphereZero}(n)) = c.$$

7 Lean obligations and naming key

`AsectionSlice` The slice/Moebius projection. This name is clean and final.

`AsectionEquivariant`
The actual octonionic round-trip functor $H_1 \to H_1$.

`AsectionFunctor` The current name of the projective groupoid-valued indexing diagram, not the octonionic round trip. A name such as `AsectionActionDiagram` would expose its role.

`TotalA` The current Grothendieck carrier. It remains provisional until total-level naturality is green.

`totalInput/totalOutput`
Current object functions. They must be promoted to total input/output functors.

Exact Gate 4 certificate.

```
AsectionTotalInput A  : A.TotalA --> H1
AsectionTotalOutput A : A.TotalA --> H1

AsectionTotalInput A ; AsectionEquivariant A =
  AsectionTotalOutput A
```

The theorem machinery remains gated until these declarations and their projective/GPV arrow naturality compile.

Arrow-level diagnostic.

Anything that must act on arrows must be checked to actually act on arrows.

A single endpoint does not retain a continuous lift, and an object function does not retain a functor. Both errors erase the arrow dimension while leaving an object-level term that can still type-check.